Working

The treated water from the sedimentation tank is allowed into the inlet chamber. It is uniformly distributed over the sand bed to a depth of 1 to 1.5 m without any disturbance. The water percolates through the filter media and gets filtered. Now the water enters the base material and comes out as filtered water. It gets collected in the laterals and discharged into the central main covered drain. Then it is finally discharged into the filtered water well. The standard rate of filtration (100 to 200 litres/m^2/hour) is continued until the difference between the water levels in the filter and the outlet chamber is slightly less than the depth of water above the sand or the loss of head reaches 0.7 to 1.2 m.

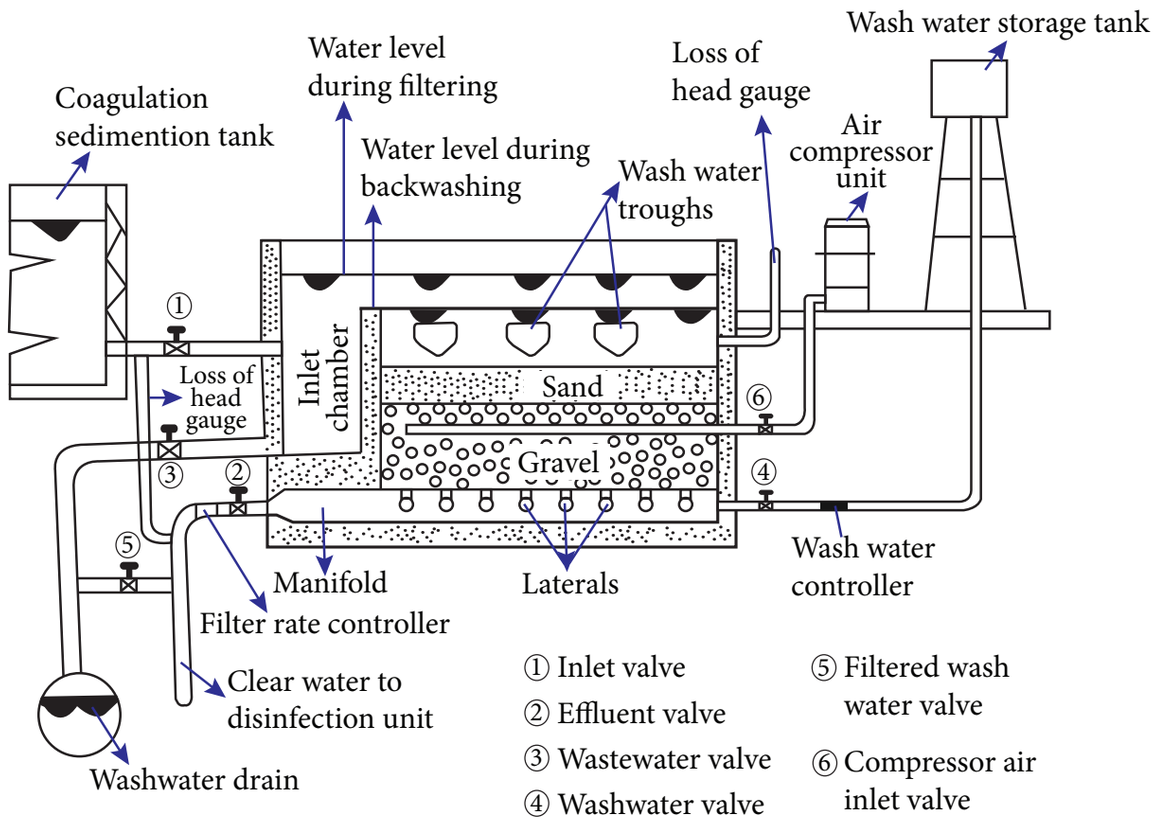
Cleaning

During working of the filter if the loss of head reaches the permissible limit, the filtration is stopped. About 20 to 30 mm sand is scraped from the top of the filter bed. The top surface is finally raked, roughened, cleaned and washed with pure water till the sand depth reduces to 0.4 m or so. Then more clean sand is added to have a minimum sand depth of about 0.45 m. The interval of cleaning may vary from 1 to 3 months.

After every cleaning, the initial filling is done by admitting filtered water from the bottom till it rises about 0.8 m above the sand. Then the fresh water is allowed to enter from the top.

2. Rapid sand filters

In rapid sand filters, the yield is about 30 times the yield given by slow sand filters for the same filter area. This is achieved by increasing the size of sand. They are also known as mechanical sand filters.



Longitudinal section of a rapid sand filter

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